

Lucas Taesung Cho

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Experience:

Tesla, Engineering Program Manager Intern (Construction), Austin TX September 2025- December 2025

Led designs with multiple disciplines for new and existing construction in a timely manner.

- Leading development of an application to reduce project start up workload by **35%** companywide through automation.
- Advised design trade-offs optimizing constructability and safety on **8 projects simultaneously** from multiple programs.
- Enabled cross functional **communication and technical ownership** between manufacturing, design, and construction.
- Presented detailed data center design analysis to directors breaking down cooling, electrical, and network systems.
- *Highlights: Gained insight into leading construction design processes for MEP and CSA*

Tesla, Mechanical Operations & Design Intern, Fremont CA January 2025 – September 2025

Designed and implemented supporting systems for the assembly line. Developed projects and led contractors and maintenance techs. Collaborated with the engineering maintenance team to troubleshoot and fix critical equipment and systems.

- Developed 4 new cooling systems supporting the assembly line using **Revit**, recycling existing equipment.
- Lead **16 projects**, overseeing system design, and coordinating with maintenance and contractors.
- Achieved **\$300,000 in cost savings** by executing projects in-house and creating cost effective solutions and designs.
- Built and troubleshooted multiple control panels for auxiliary equipment on manufacturing lines.
- *Highlights: Operated within engineering, design, and technician teams learning skills from each.*

Calpine Delta Energy Center | Power Operations Engineering Intern, Pittsburgh CA May 2024 – August 2024

Collaborated with operators and engineers on plant maintenance and project-driven enhancements to improve the plant

- Created a database of over **200 PSVs** with full identification details to forecast service and test dates.
- Calculated the total volume of liquid within the water treatment pipelines to identify critical system risks.
- Worked alongside operators, engineers, and contractors aiding in **15+ engineering projects** for a combined cycle plant.
- *Highlights: Assisted in the cold start-up of the plant under the supervision of a control room operator.*

Training Ship Golden Bear | 3/C Engineering Cadet, Vallejo CA May 2022-July 2022

Monitored and performed maintenance on a live multi-generator diesel ship under the guidance of a licensed engineer.

- Collaborated with a team of four on **20+ engineering projects** focused on pumps, electricity, filters, and plumbing.
- Produced technical drawings for key power plants and diesel engine systems.
- *Highlights: Assisted engineer in creating a response to an emergency failure in the black water system & electrical bus.*

Education: California State University, Maritime Academy, Vallejo, CA GPA: 3.62

Bachelor of Science in Mechanical Engineering – Minor in Mathematics

Prosthetic Hand Senior Project August 2024-May 2025

Lead a team to create stretchable circuits to analyze a silicone array of sensor data to control and design a prosthetic

- Utilized a **pyboard** to create a live heat map of a pressure sensor array minimizing wiring through a row-column wiring.
- Worked alongside Lawrence Berkley National Laboratory to design, manufacture, and calibrate **silicone liquid metal strain gages** with 4 sensors per finger to actuate the hand with an Ai model.
- Used **SolidWorks** to 3D model supporting fixtures to test electrical and mechanical systems of the prosthetic hand.
- Developed **PID-controlled** program to acuate finger and wrist joint movements in response to array sensor feedback
- Optimized sensor manufacturing with designed spin coater and CNC mold to enhance sensitivity and mass production.

Generator Design Competition August 2024 - December 2024

Designed and manufactured a three-phase radial flux generator with 8 pole and 9 teeth, competing against peers

- Leveraged SolidWorks and a **Prusa 3D printer** to refine tolerances, maximize winding capacity and performance
- Achieved **3rd** overall in power generation against engineering class producing **25W** within a 10 m/s wind tunnel

Highlights: Created the smallest and most material-efficient generator in the competition

Mechanical Courses: Mechatronic System Design, Fluids, Dynamics, Machinery Design, Mechanical Design, and Vibrations

Completed Courses: Calculus 3, Heat Transfer, Thermodynamics, Physics 2, Linear Algebra, Computer Aided Engineering, ENGR Computer Programming, Differential equations, Circuits, Instrumentation to Measurement Systems, Diesels, Plant operations, Manufacturing Processes 1 & 2, Electromechanics Machinery, and Experimental Methods.

Skills & Awards:

- Software: Python, C++, MATLAB, Simulink SolidWorks modeling & FEA, ANSYS, Revit, AutoCAD, and MS Office Package
- Manufacturing: CNC, Manual Lathe, Manual mill, Welding, and Soldering
- Awards: Presidents and Deans list recipient, Eagle Scout, President's Volunteer Service Award (Gold)